Keywords: sequential Monte Carlo • particle filters • differentiable resampling • variational inference

Differentiable Systematic Resampling for Variational Sequential Monte Carlo

A Sigmoid Softening of the CDF Ancestor Assignment Recovers Systematic Resampling's Variance Advantage While Allowing Full Gradient Flow

By Fredrik Cumlin & Saikat Chatterjee

arXiv:2610.12094 • NeurIPS 2026 • October 2026

Systematic resampling is the most variance-efficient particle-filter resampling scheme, but its step-function CDF blocks gradient flow. Differentiable Systematic Resampling (DSR) softens the CDF-level ancestor assignment with sigmoid functions, sorts particles via a Morton space-filling curve to preserve the banded ancestor structure, and proves that the bias vanishes exponentially in the inverse temperature—without iterative solvers.

AT A GLANCE

Problem: Systematic resampling blocks gradient flow through variational SMC due to its step-function CDF.

Why it matters: VSMC is a powerful generative modelling tool; its ELBO cannot be fully optimised without differentiable resampling.

Method: Sigmoid CDF softening with Morton-curve particle sorting; bias proven to decay exponentially in $1/\tau$.

Result: Comparable or better filtering and dynamics-learning performance vs. OT-based methods, 3–5× faster per step.

Evidence: Lorenz-96 stochastic dynamics, IAM handwriting dataset; ablation confirms Morton sorting is necessary.

The Big Picture

Sequential Monte Carlo (SMC) approximates temporal posterior distributions with weighted particles $\{(w_t^k, x_t^k)\}_{k=1}^K$. Resampling—selecting ancestors for the next particle generation—prevents degeneracy but introduces a discontinuity that blocks gradients. Variational SMC (VSMC) maximises an ELBO over proposal and model parameters via backpropagation through the full particle trajectory, so differentiable resampling is a key missing ingredient.

Systematic resampling is preferred in practice because it has $O(K^{-2})$ variance in the sample count (vs. $O(K^{-1})$ for multinomial) and is $O(K)$ to compute. DSR makes it differentiable without sacrificing these advantages.

Why Existing Approaches Fall Short

Multinomial resampling (Gumbel-softmax variants) is differentiable but has higher variance; the systematic resampling advantage is lost.

OT-based differentiable resampling preserves the systematic resampling's CDF-ordered structure but requires an iterative Sinkhorn solver at every resampling step, adding substantial overhead.

Stop-gradient resampling avoids differentiating through resampling but biases the ELBO gradient estimator systematically.

Core Mechanism

Standard systematic resampling draws $U \sim \text{Uniform}(0, 1/K)$ and assigns ancestor $a^j = \min\{i : W_{\le i} \ge (j-1)/K + U\}$, where $W_{\le i} = \sum_{l \le i} w^l$.

DSR replaces the step function with a sigmoid:

$$\tilde{A}_{ij} = \sigma\left(\frac{W_{\le i} - (j-1)/K - U}{\tau}\right) - \sigma\left(\frac{W_{\le i-1} - (j-1)/K - U}{\tau}\right)$$

The soft-resampled particle is $\tilde{x}^j = \sum_i \tilde{A}_{ij}\, x^i$.

A Morton (Z-order) space-filling curve sorts particles before resampling so that spatially proximate particles are adjacent in CDF order, preserving the *banded* ancestor matrix property that gives systematic resampling its variance efficiency.

Technical Formulation

Let $\{w^k\}_{k=1}^K$ be normalised weights with prefix sums $W_{\le k} = \sum_{j \le k} w^j$.

The soft ancestor matrix is:

$$\tilde{A}_{ij}(\tau) = \sigma\left(\tfrac{W_{\le i} - \frac{j-1}{K} - U}{\tau}\right) - \sigma\left(\tfrac{W_{\le i-1} - \frac{j-1}{K} - U}{\tau}\right)$$

Bias bound: There exist constants $C, c > 0$ such that

$$\|\mathbb{E}[\tilde{A}(\tau)] - \mathbb{E}[A]\|_\infty \le C e^{-c/\tau}$$

This exponential rate is proved from the pointwise approximation error of the sigmoid to the Heaviside step function.

Learning or Inference Procedure

Variational SMC with DSR:

1. Propagate: $x_t^k \sim q_\phi(x_t \mid x_{t-1}^k, y_t)$.
2. Weight: $w_t^k \propto p_\theta(y_t|x_t^k)\, p_\theta(x_t^k|x_{t-1}^k) / q_\phi(x_t^k|x_{t-1}^k, y_t)$.

3. Sort particles via Morton curve.
4. Apply DSR to obtain $\{\tilde{x}_t^k\}$.
5. Accumulate ELBO: $\mathcal{L} = \mathbb{E}[\log \frac{1}{K} \sum_k w_T^k]$.
6. Backpropagate through soft ancestor assignments to update ϕ, θ.

What the Guarantee Says

The bias bound $O(e^{-c/\tau})$ is stronger than the $O(\tau)$ or polynomial rates of OT-based alternatives. At temperature $\tau = 0.01$, the bias is essentially zero for practical purposes, while gradients remain well-defined through the sigmoid.

Experimental Findings

- On **Lorenz-96 stochastic dynamics**, DSR matches OT-based differentiable resampling on filtering NLL with **3–5× lower** per-step wall-clock time.
- On **IAM handwriting generation**, DSR improves log-likelihood over both multinomial and OT-based resampling.
- The absence of iterative solvers is confirmed as the primary source of the speed advantage.

Ablations and Interpretation

Without Morton sorting: The soft ancestor matrix loses its banded structure; gradient variance increases and ELBO estimates degrade.

Temperature sensitivity: An optimal τ^* balances gradient noise (low τ) vs. bias (high τ); a short sweep identifies it.

Number of particles: The advantage over multinomial resampling grows with K, consistent with the $O(K^{-2})$ vs. $O(K^{-1})$ variance rates.

Stop-gradient comparison: Stop-gradient introduces a systematic ELBO bias that DSR eliminates.

Reference

Fredrik Cumlin, Saikat Chatterjee. **Differentiable Systematic Resampling for Variational Sequential Monte Carlo**. NeurIPS 2026. arXiv:2610.12094, October 2026.
https://arxiv.org/abs/2610.12094